Complex Variables

Compiled Class Notes

21 June 2026 – 19 September 2026

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1. Complex Numbers: Basics and Arithmetic

Class of 21/06/2026

Number systems: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. In the plane, a 2D ordered pair (a, b) represents a complex number; an ordered triple (x, y, z) is a vector in 3D, and so on to 4D.

Why we need $\mathbb{C}$: $x^2 - 1 = 0$ has real roots $x = \pm 1$, but $x^2 + 1 = 0$ has no real roots; we define $x = \pm i$, where $i^2 = -1$.

Addition

For $z_1 = a_1 + i\,b_1$ and $z_2 = a_2 + i\,b_2$:

$$z_1 + z_2 = (a_1 + a_2) + i\,(b_1 + b_2)$$

Multiplication

For $z_1 = a + ib$ and $z_2 = c + id$:

$$z_1 z_2 = (a + ib)(c + id) = (ac - bd) + i\,(bc + ad)$$

Division

Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{z_1}{z_2} = \frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{c^2 + d^2} = A + iB$$

Modulus and distance

$$z = a + ib \;\Rightarrow\; |z| = \sqrt{a^2 + b^2} \qquad\qquad |z_1 - z_2| = \sqrt{(a - c)^2 + (b - d)^2}$$

Triangle inequality: in a triangle with sides a, b, c we have $a + b \geq c$. Geometrically z_1, z_2 and $z_1 + z_2$ (parallelogram rule) form a triangle, so:

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

2. Polar Form and Euler's Formula

Classes of 21/06 – 05/07/2026

Writing $z = x + iy$ with $x = r\cos\theta$ and $y = r\sin\theta$ ($r = |z|$, θ = argument):

$$z = x + iy = r\cos\theta + i\,r\sin\theta = r(\cos\theta + i\sin\theta) = r\,e^{i\theta}$$

Useful values: $i = e^{i\pi/2}$, $-1 = e^{i\pi}$. Solving $x^4 = 1$ gives the four roots $1, -1, i, -i$ (equally spaced around the unit circle, $360°/4 = 90°$ apart). Likewise $x^2 = -1 = e^{i\pi}$ gives $x = e^{i\pi/2} = i$.

Neighbourhoods

- $|z| = 1$: points *on* the unit circle
- $|z| < 1$: points *inside* the circle
- $|z| \leq 1$: points on or inside the circle
- $|z - z_1| = 1$: circle with centre z_1
- For $\delta, \varepsilon > 0$: $|z - z_0| \leq \delta$ is a closed disc about z_0; $0 < |z - z_0| \leq \delta$ is the same disc with its centre removed (deleted neighbourhood).

Roots of a complex number — Problem 1.29(b)

Find all values of z with $z^4 = -2\sqrt{3} - 2i$.

Modulus and argument of the right-hand side:

$$|z^4| = \sqrt{(2\sqrt{3})^2 + 2^2} = \sqrt{16} = 4, \qquad \tan\theta_0 = \left|\frac{y}{x}\right| = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \theta_0 = \frac{\pi}{6}$$

Since x and y are both negative, the point is in the third quadrant, so θ = π + π/6 = 7π/6:

$$z^4 = 4\,e^{i\,7\pi/6}, \qquad z^4 = r\,e^{i(\theta + 2k\pi)}$$

$$z = \sqrt{2}\;e^{i\left(\frac{7\pi/6 + 2k\pi}{4}\right)} = \sqrt{2}\;e^{i\left(\frac{7\pi}{24} + \frac{k\pi}{2}\right)}, \qquad k = 0, 1, 2, 3$$

k	Root	Argument of root
0	z1 = √2 e^(iθ)	7π/24
1	z2	7π/24 + π/2
2	z3	7π/24 + π
3	z4	7π/24 + 3π/2

The four roots lie on a circle of radius √2, spaced 90° apart.

3. Complex Functions as Mappings (z-plane → w-plane)

Classes of 05/07 – 19/07/2026 • Book 2 (Brown), p. 33

A complex function w = f(z) maps points of the **z-plane** (z = x + iy) to the **w-plane** (w = u + iv).

Example: w = z^2

$$w = z^2 = (x + iy)^2 = (x^2 - y^2) + 2xy\,i \Rightarrow u = x^2 - y^2, \quad v = 2xy$$

Lines u = c_2 > 0 and v = c_1 > 0 in the w-plane (vertical and horizontal lines) correspond to families of hyperbolas $x^2 - y^2 = c_2$ and $2xy = c_1$ in the z-plane (the curve xy = const. has branches in the 1st and 3rd quadrants).

Elementary transformations (19/07/2026)

- **Translation:** w = z + 1, or in general w = z + (a + ib) shifts every point by the fixed vector (a, b).
- **Rotation:** w = iz. Since $i = e^{i\pi/2}$ and $z = re^{i\theta}$, we get $w = r\,e^{i(\theta + \pi/2)}$, a rotation by 90° anticlockwise.
- **Reflection:** z = x + iy ↦ $\bar{z}$ = x − iy reflects the point in the real axis.
- **Squaring:** w = z^2 (above).

4. Harmonic Functions and the Cauchy–Riemann Equations

Class of 26/07/2026 • Book 2, p. 80, Ex. 9

Laplacian:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = u_{xx} + u_{yy} + u_{zz}$$

Two physical equations (u = u(x, t)):

- **Heat equation:** $\partial u/\partial t = \nabla^2 u$, i.e. $u_t = u_{xx}$ (in one space dimension).
- **Wave equation:** $\partial^2 u/\partial t^2 = \nabla^2 u$, i.e. $u_{tt} = u_{xx}$.

Harmonic function

If $\nabla^2 u = 0$, then u satisfies **Laplace's equation** and is called **harmonic**.

Cauchy–Riemann (C–R) equations for f(z) = u + iv

$$u_x = v_y, \qquad u_y = -v_x$$

Example: $w = z^2$

$u + iv = (x + iy)^2 = x^2 - y^2 + 2xy\,i$, so $u = x^2 - y^2$ ($u_x = 2x$, $u_y = -2y$) and $v = 2xy$ ($v_x = 2y$, $v_y = 2x$). Hence $u_x = v_y$ and $u_y = -v_x$: the C–R equations are satisfied.

Worked problem: harmonic conjugate

Show that (a) $u = x^2 - y^2$ is harmonic; (b) find its harmonic conjugate v; (c) express w = f(z) as a function of z only.

(a) $u_{xx} = 2$, $u_{yy} = -2$, so $u_{xx} + u_{yy} = 2 - 2 = 0$. Laplace's equation holds, so u is harmonic.

(b) If v is the harmonic conjugate of u, then by C–R: $v_y = u_x = 2x$ and $v_x = -u_y = -(-2y) = 2y$. (Also $v_{xx} = 0$ and $v_{yy} = 0$, so $v_{xx} + v_{yy} = 0$ and v is harmonic.) Using the differential

$$dv = v_x\,dx + v_y\,dy = 2y\,dx + 2x\,dy = 2\,(y\,dx + x\,dy) = 2\,d(xy)$$

Integrating: **$v = 2xy + C$**.

(c) $w = f(z) = u + iv = x^2 - y^2 + 2xy\,i + iC$. *Side note:* if $f(z) = u(x, y) + i\,v(x, y)$, then putting $x = z$, $y = 0$ gives $f(z) = u(z, 0) + i\,v(z, 0)$. Here this yields

$$f(z) = z^2 + C$$

(with C a constant).

Another example: $w = z^3$

$$(x + iy)^3 = x^3 + 3x^2(iy) - 3xy^2 - iy^3 = (x^3 - 3xy^2) + i\,(3x^2y - y^3)$$

so $u = x^3 - 3xy^2$ and $v = 3x^2y - y^3$.

5. Complex (Contour) Integration

Class of 16/08/2026 • Book 1, p. 119, Ex. 4.2

A contour integral is a line integral along a path C from A to B, in the same way that **work** is a line integral of force:

$$W = \mathbf{F} \cdot \mathbf{d} = F\,d\cos\theta \;\longrightarrow\; W = \int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot d\mathbf{r}$$

Parametrising a circle |z| = 2:

$$z = 2e^{i\theta}, \qquad dz = \tfrac{dz}{d\theta}\,d\theta = 2i\,e^{i\theta}\,d\theta, \qquad \bar{z} = 2e^{-i\theta}$$

Problem: evaluate $\int_C \bar{z}\,dz$ (a) along C: $z = t^2 + ti$ (a parabola from 0 to 4 + 2i); (b) along C: the line from 0 to 2i, then to 4 + 2i.

(a) Parabola

$z = t^2 + ti$, so $\bar{z} = t^2 - ti$ and $dz = (2t + i)\,dt$, with t from 0 to 2:

$$\int_0^2 (t^2 - ti)(2t + i)\,dt = \int_0^2 (2t^3 + t - t^2 i)dt = \left[\tfrac{t^4}{2} + \tfrac{t^2}{2} - \tfrac{t^3}{3}\,i\right]_0^2 = 10 - \tfrac{8}{3}\,i$$

Answer (a): 10 − (8/3) i

(b) Two straight segments, C = C_1 + C_2

Along C_1 (the imaginary axis, 0 to 2i): z = yi, dz = i dy, $\bar{z}$ = −yi, so $\bar{z}$ dz = y dy:

$$\int_{C_1} \bar{z}\,dz = \int_0^2 y\,dy = \left[\frac{y^2}{2}\right]_0^2 = 2$$

Along C_2 (the line y = 2, x from 0 to 4): $\bar{z}$ = x − 2i, dz = dx:

$$\int_{C_2} \bar{z}\,dz = \int_0^4 (x-2i)\,dx = 8 - 8i$$

Adding: $\int_C \bar{z}\,dz$ **= 2 + (8 − 8i) = 10 − 8i**. The two paths give different answers because $\bar{z}$ is not analytic.

6. Cauchy's Integral Theorem and Integral Formula

Classes of 16/08 – 31/08/2026

Cauchy's Integral Theorem

$$\oint_C f(z)\,dz = 0$$

when f is analytic inside and on the closed contour C.

Exam suggestion noted in class: roots of complex numbers, harmonic function, analytic function, conjugate (Chapters 1 & 3).

Cauchy's Integral Formula (23/08/2026)

$$\oint_C \frac{f(z)}{z-a}\,dz = 2\pi i\, f(a)$$

for a lying inside C and f analytic on and inside C.

Example (p. 169, Ex. 6)

Evaluate $\oint_C$ f(z)/(z + i) dz with f(z) = z / (9 − z^2) and C: |z| = 2. The point z = −i lies inside C (while z = ±3 lies outside), so apply the formula with a = −i:

$$f(-i) = \frac{-i}{9-(-i)^2} = \frac{-i}{10}, \qquad \oint_C \frac{f(z)}{z-(-i)}\,dz = 2\pi i\, f(-i) = 2\pi i\left(\frac{-i}{10}\right) = \frac{2\pi}{10} = \frac{\pi}{5}$$

Answer: π/5

Using partial fractions (Ex. on p. 235; 31/08/2026)

Evaluate $\oint_C$ (5z − 2) / [z(z − 1)] dz, where C encloses both z = 0 and z = 1.

Split into partial fractions:

$$\frac{5z-2}{z(z-1)} = \frac{-2}{z(-1)} + \frac{5(1)-2}{1\,(z-1)} = \frac{2}{z} + \frac{3}{z-1}$$

Apply Cauchy's formula to each term (f ≡ 1 in each case):

$$\oint_C f(z)\,dz = 2\pi i \cdot 2 + 2\pi i \cdot 3 = 10\pi i$$

Answer: 10πi

Same integral by residues. Last class's method: $\text{Res}_{z=0}$ f = 2 and $\text{Res}_{z=1}$ f = 3, so $\oint$ f dz = 2πi Σ Res = 2πi (2 + 3) = 10πi.

Check via the residue at infinity. Form $(1/z^2)\, f(1/z)$:

$$\frac{1}{z^2} f\left(\frac{1}{z}\right) = \frac{1}{z^2} \cdot \frac{5/z - 2}{(1/z)(1/z - 1)} = \frac{5 - 2z}{z(1 - z)}$$

Its residue at z = 0 is 5, so $\oint f\, dz = 2\pi i \cdot 5 = 10\pi i$, agreeing with the answer above.

A pole outside the contour

For $f(z) = 1 / [z\,(z - 2)^4]$ on a circle around z = 2 that does **not** contain z = 0 (z = 0 is not within the contour), only the pole of order 4 at z = 2 counts:

$$\mathrm{Res}_{z=2} f = \frac{1}{3!} \lim_{z \to 2} \frac{d^3}{dz^3}\left(\frac{1}{z}\right) = \frac{1}{6}\left(\frac{-6}{z^4}\right)_{z=2} = -\frac{1}{16}$$

$$\oint_C f(z)\, dz = 2\pi i\left(-\frac{1}{16}\right) = -\frac{\pi i}{8}$$

Handy series for expansions

Geometric progression: $1 + r + r^2 + \ldots + r^{n-1} = (1 - r^n)/(1 - r)$, which for $|r| < 1$ tends to $1/(1 - r)$.

Binomial series (from Pascal's triangle): rows read *horizontally* give the coefficients of $(1 + x)^n$ (positive powers); the *diagonals* give the series for negative powers:

$$(1-x)^{-1} = 1 + x + x^2 + \cdots, \quad (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \cdots, \quad (1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \cdots$$

7. Series: Taylor and Laurent

Classes of 31/08 – 12/09/2026

Taylor series

If f is analytic in $|z - a| < R$, then in that disc:

$$f(z) = f(a) + f'(a)(z-a) + \frac{f''(a)}{2!}(z-a)^2 + \frac{f'''(a)}{3!}(z-a)^3 + \cdots + \frac{f^{(n)}(a)}{n!}(z-a)^n + \cdots$$

Example: expand f(z) = ln(1 + z) about z = 0

$$f(z) = \ln(1+z),\ f'(z) = \tfrac{1}{1+z},\ f''(z) = \tfrac{-1}{(1+z)^2},\ f'''(z) = \tfrac{2}{(1+z)^3},\ \ldots,\ f^{(n+1)}(z) = (-1)^n\, n!\ (1+z)^{-(n+1)}$$

$$f(0) = 0, \quad f'(0) = 1, \quad f''(0) = -1, \quad f'''(0) = 2$$

$$\ln(1+z) = f(0) + f'(0)\, z + \frac{f''(0)}{2!} z^2 + \frac{f'''(0)}{3!} z^3 + \cdots = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots$$

The series converges when $|z - 0| < 1$.

Laurent series

A Laurent series is like a Taylor series but it can also include **negative** powers of (z − a). The Laurent series of f(z) about z = a has the form:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n = \cdots + \frac{a_{-2}}{(z-a)^2} + \frac{a_{-1}}{z-a} + a_0 + a_1(z-a) + a_2(z-a)^2 + \cdots$$

- **Principal part:** the terms with negative powers, $a_{-1}/(z - a) + a_{-2}/(z - a)^2 + \ldots$
- **Analytic part:** the terms $a_0 + a_1(z - a) + a_2(z - a)^2 + \ldots$

Singular point: $f(z) = 1/(z - 2)$ is not analytic at z = 2; such a point is called a **singular point**.

Example 1: f(z) = 1 / [z(1 − z)] about z = 0

Singular points: z = 0 and z = 1. Two regions:

For |z| < 1 (0 < |z| < 1):

$$\frac{1}{1-z} = (1-z)^{-1} = 1 + z + z^2 + z^3 + \cdots \Rightarrow \frac{1}{z(1-z)} = \frac{1}{z} + 1 + z + z^2 + z^3 + \cdots$$

For |z| > 1 (so |1/z| < 1):

$$\frac{1}{z(1-z)} = -\frac{1}{z^2}\left(1 - \frac{1}{z}\right)^{-1} = -\frac{1}{z^2}\left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right) = -\frac{1}{z^2} - \frac{1}{z^3} - \frac{1}{z^4} - \cdots$$

The region 0 < |z| < 1 is a punctured disc, and |z| > 1 is the exterior of the unit circle (an annular region).

Example 2 (12/09/2026): Laurent series of f(z) = 1 / [(z − 1)(z + 2)] about z = 1

Let w = z − 1, so z + 2 = w + 3 and f = 1 / [w(w + 3)].

Region 0 < |z − 1| < 3 (i.e. 0 < |w/3| < 1):

$$f = \frac{1}{3w}\left[1 + \frac{w}{3}\right]^{-1} = \frac{1}{3w}\left[1 - \frac{w}{3} + \frac{w^2}{9} - \frac{w^3}{27} + \cdots\right]$$

$$f(z) = \frac{1}{3(z-1)} - \frac{1}{9} + \frac{z-1}{27} - \frac{(z-1)^2}{81} + \cdots$$

Region |z − 1| > 3 (i.e. |3/w| < 1):

$$f = \frac{1}{w^2\left(1 + \frac{3}{w}\right)} = \frac{1}{w^2}\left(1 - \frac{3}{w} + \frac{9}{w^2} - \frac{27}{w^3} + \cdots\right)$$

$$f(z) = \frac{1}{(z-1)^2} - \frac{3}{(z-1)^3} + \frac{9}{(z-1)^4} - \cdots$$

8. Residues and the Residue Theorem

Classes of 12/09 – 16/09/2026

Definition

The Laurent series of f(z) at z = a is f(z) = Σ $a_n(z - a)^n$ (n from −∞ to ∞). The **residue** of f at z = a is the coefficient a_{-1} of 1/(z − a).

Example: $e^{1/z} = 1 + 1/z + 1/(2!\, z^2) + 1/(3!\, z^3) + \ldots$, so the residue of $e^{1/z}$ at z = 0 is **1**.

Formulas for computing residues

(i) Simple pole (pole of order one) at z = a:

$$\mathrm{Res}_{z=a} f(z) = \lim_{z \to a} (z-a)\, f(z)$$

(ii) Pole of order n at z = a:

$$\mathrm{Res}_{z=a} f(z) = \frac{1}{(n-1)!} \lim_{z \to a} \frac{d^{n-1}}{dz^{n-1}} [(z-a)^n f(z)]$$

Recognising poles. If f(z) = φ(z)/(z − a) with φ(a) ≠ 0, then z = a is a simple pole. Similarly, if f(z) = φ(z)/$(z - a)^n$ with φ(a) ≠ 0, then z = a is a pole of order n.

Example: residues of f(z) = z^2 / $[(z + 1)^2(z - 2)]$ at z = −1 and z = 2

At z = 2 (simple pole):

$$\text{Res}_{z=2} f = \lim_{z\to 2}(z-2)\,f(z) = \frac{z^2}{(z+1)^2}\Big|_{z=2} = \frac{4}{9}$$

At z = −1 (pole of order 2). Let φ(z) = z^2/(z − 2), so f = φ(z)/$(z + 1)^2$:

$$\text{Res}_{z=-1} f = \frac{d}{dz}\left[\frac{z^2}{z-2}\right]_{z=-1} = \frac{z^2-4z}{(z-2)^2}\Big|_{z=-1} = \frac{5}{9}$$

(The values 4/9 and 5/9 were worked out from the formulas above; the printed numbers in the original page were only partly legible.)

9. Evaluating Real Integrals by Residues

Classes of 16/09 – 19/09/2026

(A) Trigonometric integral over [0, 2π]

Q. Evaluate $\int_0^{2\pi} d\theta / (5 - 3\cos\theta)$ using the Cauchy (residue) technique.

Substitution. Let C: |z| = 1 and $z = e^{i\theta}$ with 0 ≤ θ ≤ 2π, so $z^{-1} = e^{-i\theta}$. From Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$:

$$\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2} = \frac{z+z^{-1}}{2}, \qquad \sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i} = \frac{z-z^{-1}}{2i}$$

Also $dz = i\,e^{i\theta}\,d\theta = iz\,d\theta$, so $d\theta = dz/(iz)$. Then:

$$5 - 3\cos\theta = 5 - \frac{3}{2}\left(\frac{z^2+1}{z}\right) = \frac{10z-3z^2-3}{2z} = -\frac{(3z-1)(z-3)}{2z}$$

$$\int_0^{2\pi} \frac{d\theta}{5-3\cos\theta} = \oint_{|z|=1} \frac{dz/(iz)}{-(3z-1)(z-3)/(2z)} = \oint_{|z|=1} \frac{-2}{i(3z-1)(z-3)}\,dz$$

Let f(z) = −2 / [i(3z − 1)(z − 3)]. It has singularities at z = 3 and z = 1/3; only **z = 1/3** lies inside |z| = 1, and it is a simple pole:

$$\text{Res}_{z=1/3} f = \lim_{z\to 1/3}\left(z-\frac{1}{3}\right)\frac{-2}{i(3z-1)(z-3)} = \frac{-2}{3i(z-3)}\Big|_{z=1/3} = \frac{-2}{3i(-8/3)} = \frac{1}{4i}$$

By the Cauchy residue theorem ($\oint_C f\,dz = 2\pi i \times$ sum of residues inside C):

$$\int_0^{2\pi} \frac{d\theta}{5-3\cos\theta} = 2\pi i \cdot \frac{1}{4i} = \frac{\pi}{2}$$

Answer: π/2

(B) Improper integral over the real line

Q (19/09/2026). Using complex-variable techniques, evaluate $\int_{-\infty}^{\infty} dx / (x^4 + 1)$.

Contour. Consider $\oint_C f(z)\,dz$ where $f(z) = 1/(z^4 + 1)$ and C is the closed contour made of the real axis from −R to R together with the upper half-circle C_R: |z| = R.

Poles are given by $z^4 + 1 = 0$, i.e. $z^4 = -1$:

$$z = \cos\frac{(2n+1)\pi}{4} + i\sin\frac{(2n+1)\pi}{4}, \qquad n = 0, 1, 2, 3$$

$$z_1 = e^{i\pi/4}, \quad z_2 = e^{i3\pi/4}, \quad z_3 = e^{i5\pi/4}, \quad z_4 = e^{i7\pi/4}$$

There are four poles, but only **z_1 and z_2** lie within the contour (upper half-plane). They are simple poles:

$$\text{Res}_{z=z_1} f = \lim_{z\to z_1}(z-z_1)f(z) = \frac{1}{4z_1^3}$$

$$\text{Res}_{z_1} = \frac{1}{4e^{i3\pi/4}} = \frac{1}{4}\left(-\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right), \qquad \text{Res}_{z_2} = \frac{1}{4e^{i9\pi/4}} = \frac{1}{4}\left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right)$$

Therefore, with the residue theorem:

$$\int_{-R}^{R} f(x)\,dx + \int_{C_R} f(z)\,dz = 2\pi i\,(\text{(sum of residues)}) = 2\pi i\left(-\frac{i}{2\sqrt{2}}\right) = \frac{\pi}{\sqrt{2}}$$

The arc contributes nothing as R → ∞. On C_R, $z = Re^{i\theta}$ and $|dz| = R\,d\theta$. Using $|z_1 + z_2| \geq ||z_1| - |z_2||$, we get $|z^4 + 1| \geq R^4 - 1$, so:

$$\left|\int_{C_R} \frac{dz}{z^4+1}\right| \leq \int_{C_R} \frac{|dz|}{|z^4+1|} \leq \int_0^{\pi} \frac{R\,d\theta}{R^4-1} = \frac{\pi R}{R^4-1} \approx \frac{\pi}{R^3} \to 0 \quad (R \to \infty)$$

Result

$$\int_{-\infty}^{\infty} \frac{dx}{x^4+1} = \frac{\pi}{\sqrt{2}}$$

10. Practice Problems Listed in the Notes

Set of real integrals to be done by contour/residue methods:

$$(1)\ \int_0^{\infty} \frac{\sin x}{x}\,dx \qquad (2)\ \int_{-\infty}^{\infty} \frac{x\sin x}{x^2+2x+5}\,dx \qquad (3)\ \int_{-\infty}^{\infty} \frac{\cos mx}{x^2+1}\,dx \qquad (4)\ \int_{-\infty}^{\infty} \frac{dx}{x^6+1}$$

Also posed after the trigonometric example (16/09/2026):

$$\oint \frac{d\theta}{(5-3\cos\theta)^2}$$

Other exercises to remember: (i) for Problem 1.29(b), draw the four roots on the circle of radius √2; (ii) revise the C–R equations and harmonic conjugates for the exam.